

A generic NOVA pass recovers a DNT training set's known pharmacology

EPA/Shafer multi-well MEA network formation assay (Brown et al. 2016)
— exploratory reanalysis

2026-07-15

Headline

NOVA's standard pipeline was run over this dataset with no compound-specific tuning: normalise to vehicle controls, PCA, trajectory summary, rank by effect size. Of the 25 largest changes it surfaced, **20 are the two compounds this training set treats as active** (Mevastatin and Sodium Orthovanadate). The designated negative, Acetaminophen, appears **0 times**: it does not reach the top 25 at all. Every one of the 25 is a *decrease* relative to vehicle controls.

This is a statement about **prediction, not neurotoxicity**: the ranking recovers labels the training set already assigned, on 41 conditions across 8 metrics and 3 timepoints. Nothing here was tested.

Hero figure — actives fall, the negative does not

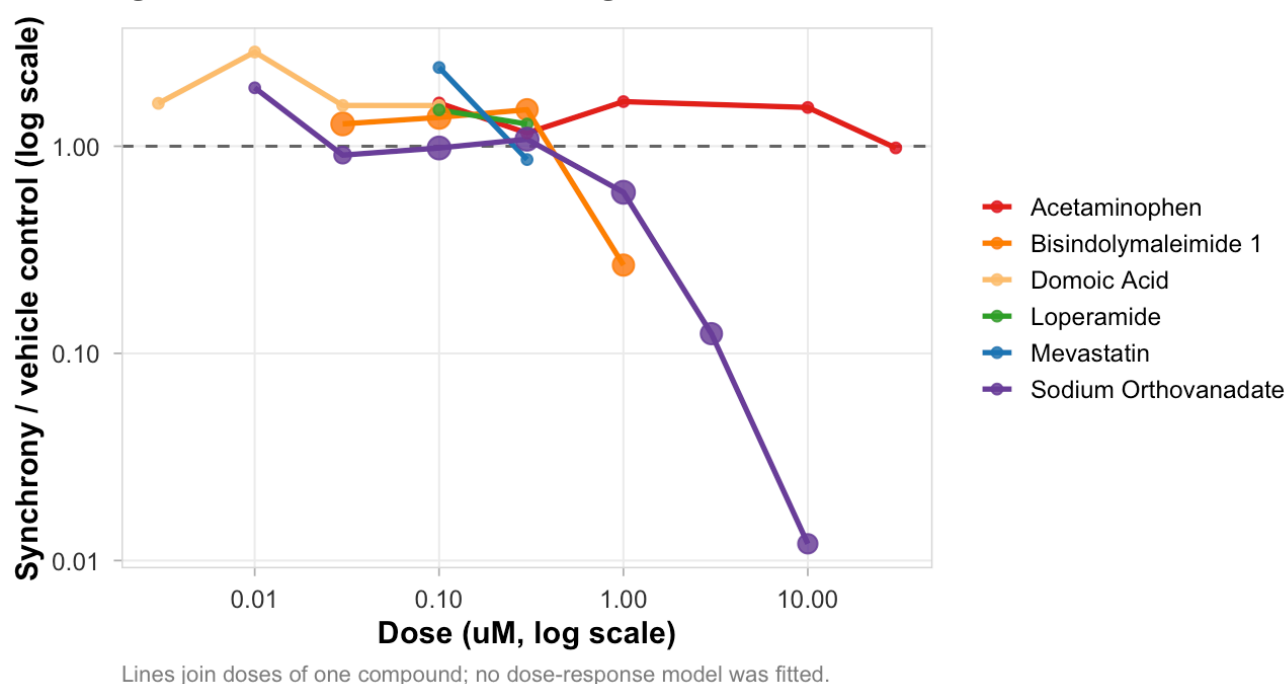


Figure 1: Synchrony index as a fraction of same-plate vehicle controls at DIV 7, by compound and dose. Dashed line = vehicle (no change). Each point is a condition's median across its replicate wells, counting a well by plate and ID; only conditions with at least 4 wells are shown. Sodium orthovanadate and mevastatin - the compounds this training set designates active - fall away with dose, as does bisindolymaleimide-1. Acetaminophen, the designated negative, tracks its controls across three orders of magnitude.

The largest changes that more than one well agrees on

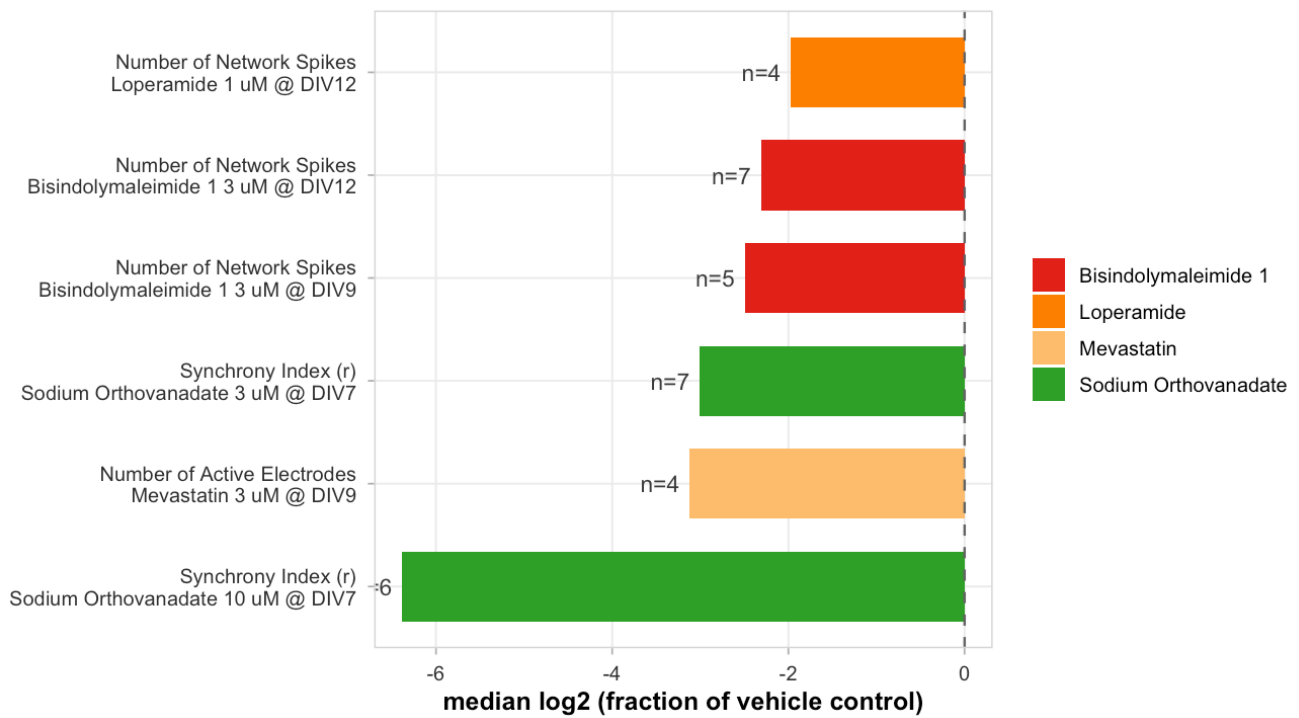


Figure 2: Top baseline-normalised changes with at least 4 replicate wells, selected by effect-size rank from findings.csv rather than by eye. Bars are median log2 fold-change vs same-plate vehicle controls; the well count is printed on each.

	Condition	Metric	DIV	vs control	Wells	Plates
5	Sodium Orthovanadate 10 uM	Synchrony Index (r)	DIV7	83.44x down	6	6
20	Mevastatin 3 uM	Number of Active Electrodes	DIV9	8.70x down	4	4
22	Sodium Orthovanadate 3 uM	Synchrony Index (r)	DIV7	8.03x down	7	6
36	Bisindolymaleimide 1 3 uM	Number of Network Spikes	DIV9	5.64x down	5	4
39	Bisindolymaleimide 1 3 uM	Number of Network Spikes	DIV12	4.96x down	7	6
46	Loperamide 1 uM	Number of Network Spikes	DIV12	3.92x down	4	4

Caveats

⚠ Warning

Exploratory. Nothing here is a tested claim.

- **Selection inflates the top of the list.** 944 condition × metric × timepoint comparisons were scanned and sorted by effect size; sorting a large family of noisy estimates puts the luckiest first. No correction is applied because nothing is a hypothesis test.
- **199 of 944 comparisons were set aside for resting on fewer than 4 wells**, and are not shown. Raw effect size cannot distinguish a large change in three wells from one in thirty: unfiltered, 14 of the top 25 rested on a **single well**, including a ~1680× effect. Ranking within what remains is still by raw effect size.
- **DIV 5 is excluded entirely.** Values are a fraction of same-plate vehicle controls, and at DIV 5 those controls are themselves silent — the control mean is exactly zero for network spikes in 4 of 6 experiments, making the ratio undefined for 63–81% of wells. Cortical networks have not formed at DIV 5; that is what this assay measures.
- **No timepoint here is pre-treatment.** Compounds are applied from the start of the 12-day culture, so “change across DIV” is not “onset of effect”.
- **Plate, batch and dose structure are not modelled.** Wells nest in plates, plates in culture dates, doses in compounds, and DIV is a repeated measure on the same wells. Conditions are treated as flat groups.
- **Prediction is not causation.** Recovering the training set’s own active/inactive labels says the assay predicts those labels. It says nothing about developmental neurotoxicity in humans.
- **The ns.* and burst-interval metrics are excluded from the state space**, because their 33–42% missingness means “this well did not burst” rather than “measurement lost”, and mean-imputing it would invent burst statistics for silent wells.

Source

US EPA (Brown et al. 2016, *Toxicol Sci* 154(1):126–139), public domain (17 U.S.C. § 105). URL, checksum and the full structural record are in `SOURCE.md` and reproduced in the HTML notebook.

Generated with NOVA 0.4.0. Discovery pass: `run_discovery.R`. Figures use `nova_theme()/nova_palette()`. Every result shown was selected by effect-size rank from `findings.csv`; none was hand-picked.